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→ **Binary Tree:**

Node can have no more than 2 sub tree.

0, 1, 2

→ **Strictly Binary tree:**

Each node has either 0 (or) 2 children

0, 2

→ **Complete Binary tree:**

Tree where no. of node at any level i, no. of nodes is $2^i$.

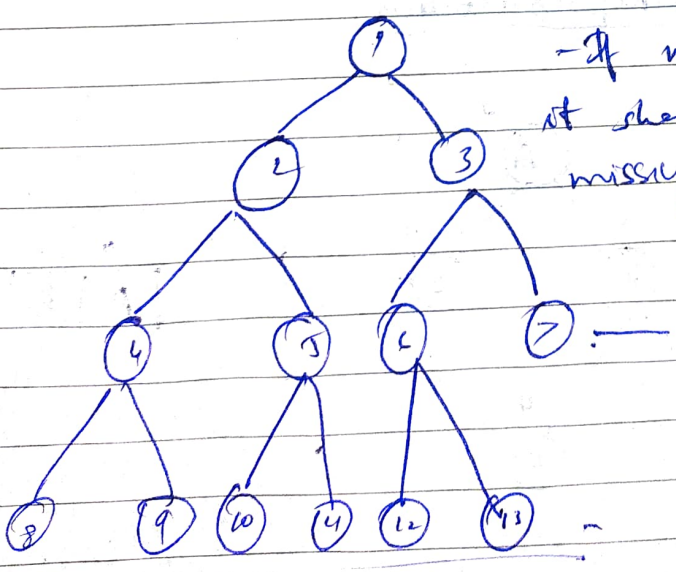
Complete B.T = Strictly Binary Tree.

Strictly B.T ≠ Complete Binary Tree.

→ **Almost complete Binary Trees / Nearly complete Binary Tree:**

Tree which is complete atleast till maximum level - 1

- Tree should complete upto second last level.




- If nodes are missing it should only be missing on right side.

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→ P.T., total number of nodes in complete binary tree of depth 'd' is $2^{d+1}-1$. Also find total number of non-leaf nodes

Sol

Total nodes at given level ⇒ $2^d$

∴ Total number of node in a complete binary tree of depth 'd'

⇒ $2^0 + 2^1 + 2^2 + \cdots 2^d$

w.k.T., sum upto 4 terms ⇒ $\frac{a(r^n-1)}{r-1}$

$r=2$, $a=2^0$, $n=d+1$.

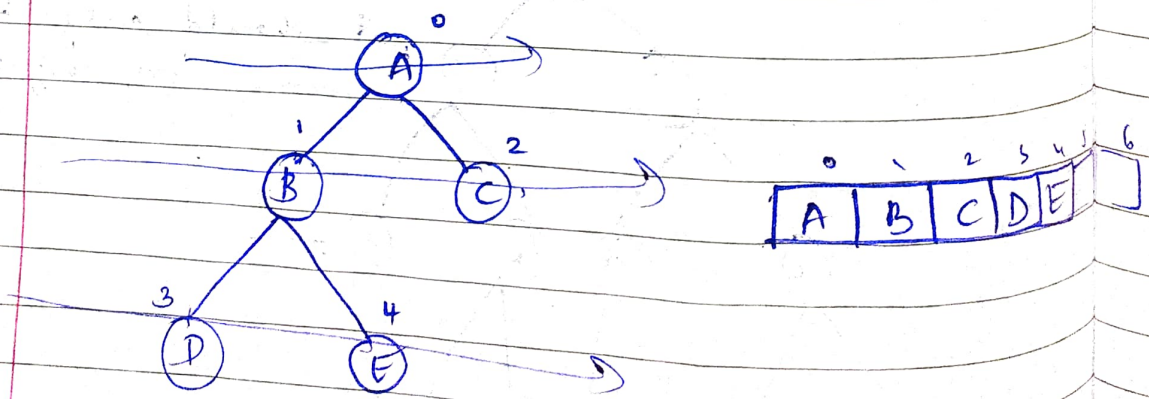
⇒ $\frac{2^0(2^{d+1}-1)}{2-1}$ ⇒ $\frac{2^{d+1}-1}{1}$ ⇒ $\boxed{2^{d+1}-1}$

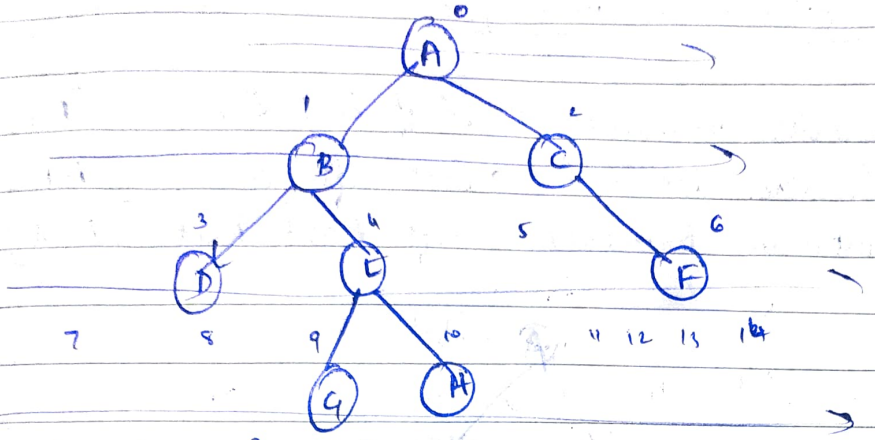
Total number of non-leaf nodes = Total - leaf

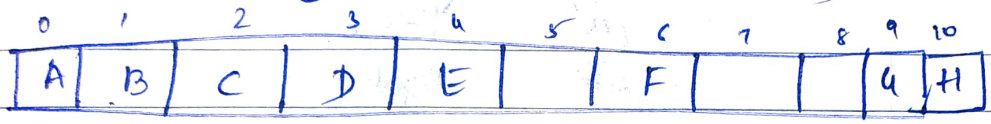
⇒ $2^{d+1}-1-2^d$

⇒ $\boxed{2^d-1}$

→ Sequential Representation:

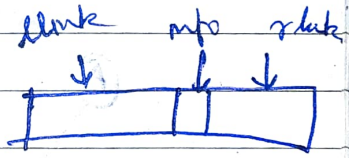


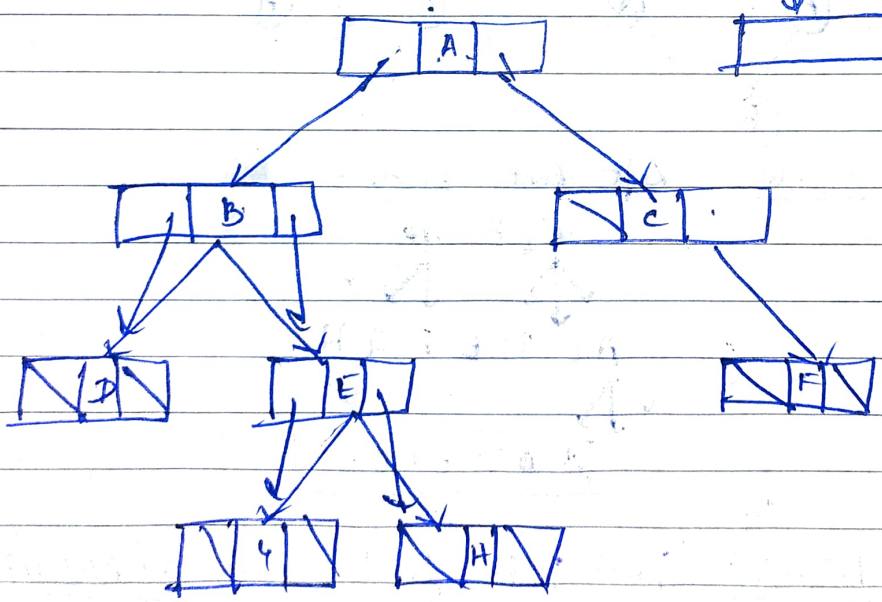



$2i+1$ → left child
$2i+2$ → Right child

→ Linked Representation:

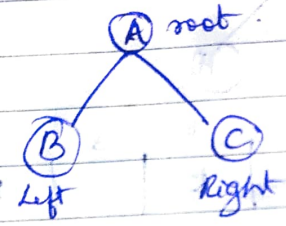


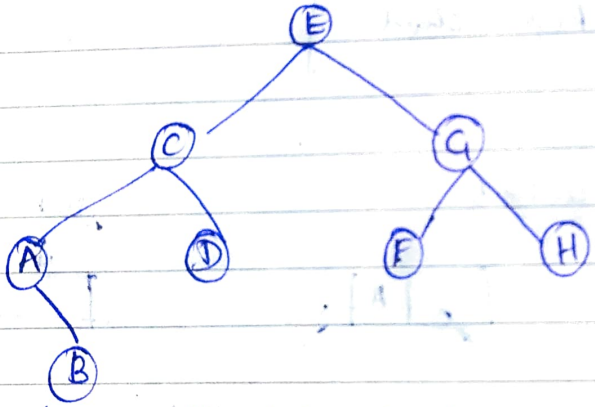

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→ Tree Traversal Techniques:

① Inorder Traversal → Left Root Right . [B A C]
② Preorder traversal → Root Left Right [A B C]
③ Post order traversal → Left Right Root [B C A]




⇒




① Inorder: Left Root Right .

C E G

A C D E F G H

A B C D E F G H

② Preorder: Root Left Right . ⇒ E C A B D G F H

E C G

E C A D G F H

E C A B D G F H

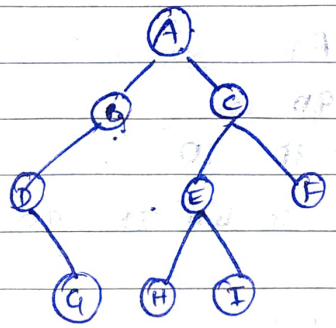
③ Postorder: Left Right Root → BADCFHGE

C* G* E

A D C F H G

BADC FHGE

②




a] Inorder: Left Root Right

B A C

D B A E C F

D G B A H E I C F

b] Preorder: Root Left Right

A B C

A B D C E F

A B D G C E H I F

c] Post order: Left Right Root

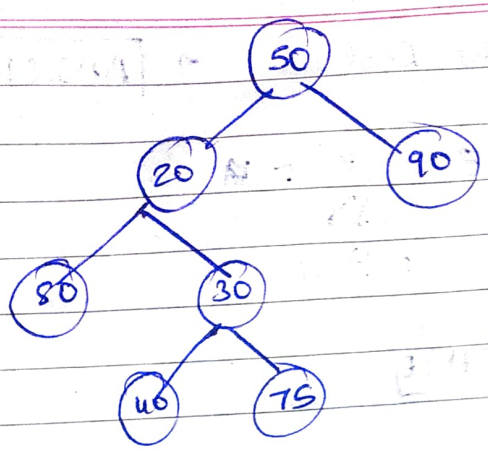
B C A

D B E F C A

G D B H I E F C A

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③




a] Inorder: L Root R

~~40 30 75 80 20~~ 20 50 90

80 20 30 50 90

80 20 40 30 75 50 90

b] Preorder: Root Lf Rgt

50 20 90

50 20 80 30 90

50 20 80 30 40 75 90

c] Postorder: Lf Ri Root

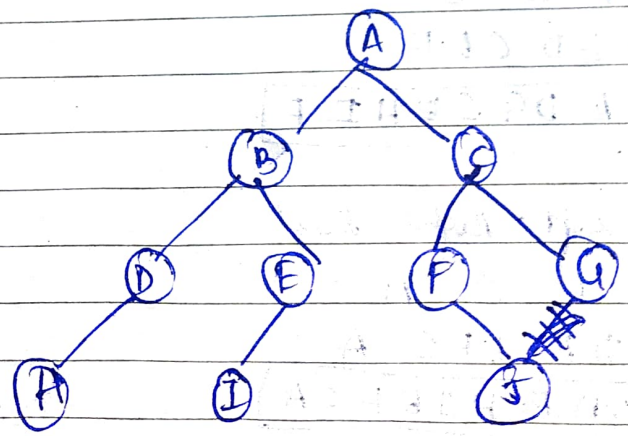
20 90 50

80 30 20 90 50

80 40 75 30 20 90 50

④

| A | B | C | D | E | F | G | H | | I | | | J | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | |




Incomplete tree

Inorder: B A C

D B E A F C G

H D B I E A F J C G

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Preorder: A B C

A B D E C F G

A B H D E I C F J G

Postorder: B C A

D E B C F G A

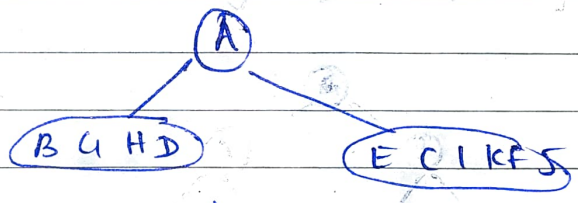
H D I E B C J F G A

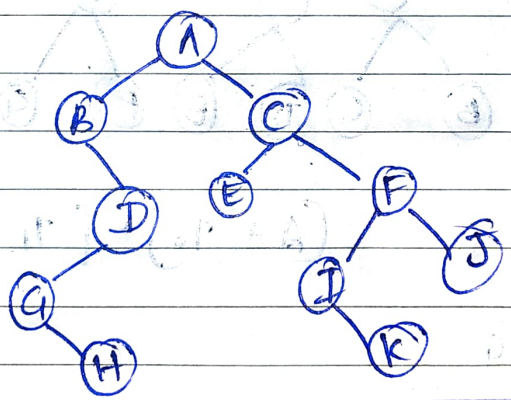
② Construction of tree from traversal:

Preorder: A B D G H C E F I K J

Inorder: B G H D A E C I K F J

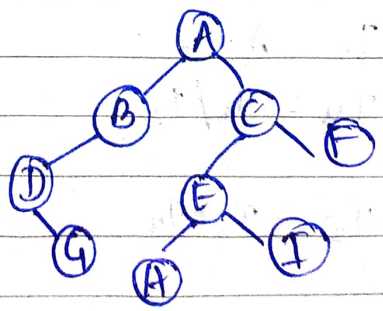
left right




② Preorder: A B D G C E H I F

Inorder: D G B A H E I C F




~~A B C~~

B A C

D B A E C F

D G B A H E I C F

(3) Postordu: 1 4 3 2 6 8 9 11 10 7 5

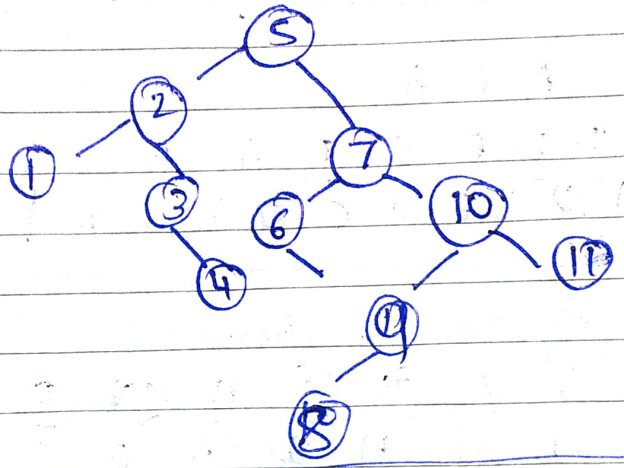
Inordu: 1 2 3 4 5 6 7 8 9 10 11

2 5 7

1 2 3 5 6 7 10

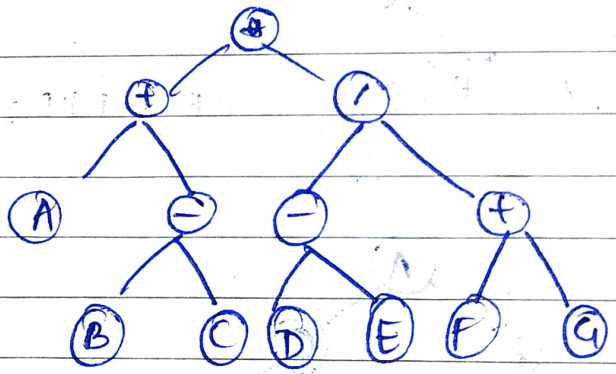
1 2 4 3 5 6 7 11 10

1 2




→ Expression Tree: [Strictly Binary tree]

[A + (B - C)] * [(D - E) / (F + G)]




Leaf ⇒ operand

root ⇒ operator

$T_0 \rightarrow B - C$

$T_1 = D - E$

$T_2 = F + G$

$[A + T_0] * [T_1 / T_2]$

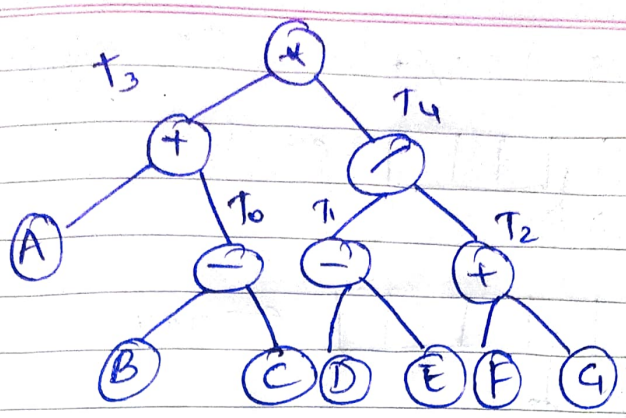
$T_3 = A + T_0$

$T_4 = T_1 / T_2$

$T_3 * T_4$

$T_5 = T_3 * T_4$

$T_5$




$(A + B * C) - ((D * E + F) / G)$

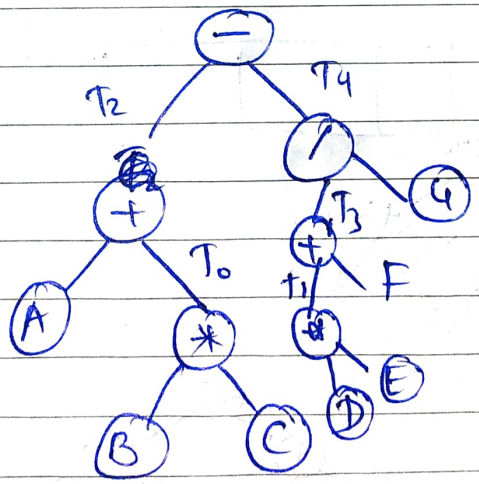
→ $T_0 = B * C$ $\quad (A + T_0) - ((T_1 + F)/G)$

$T_1 = D * E$

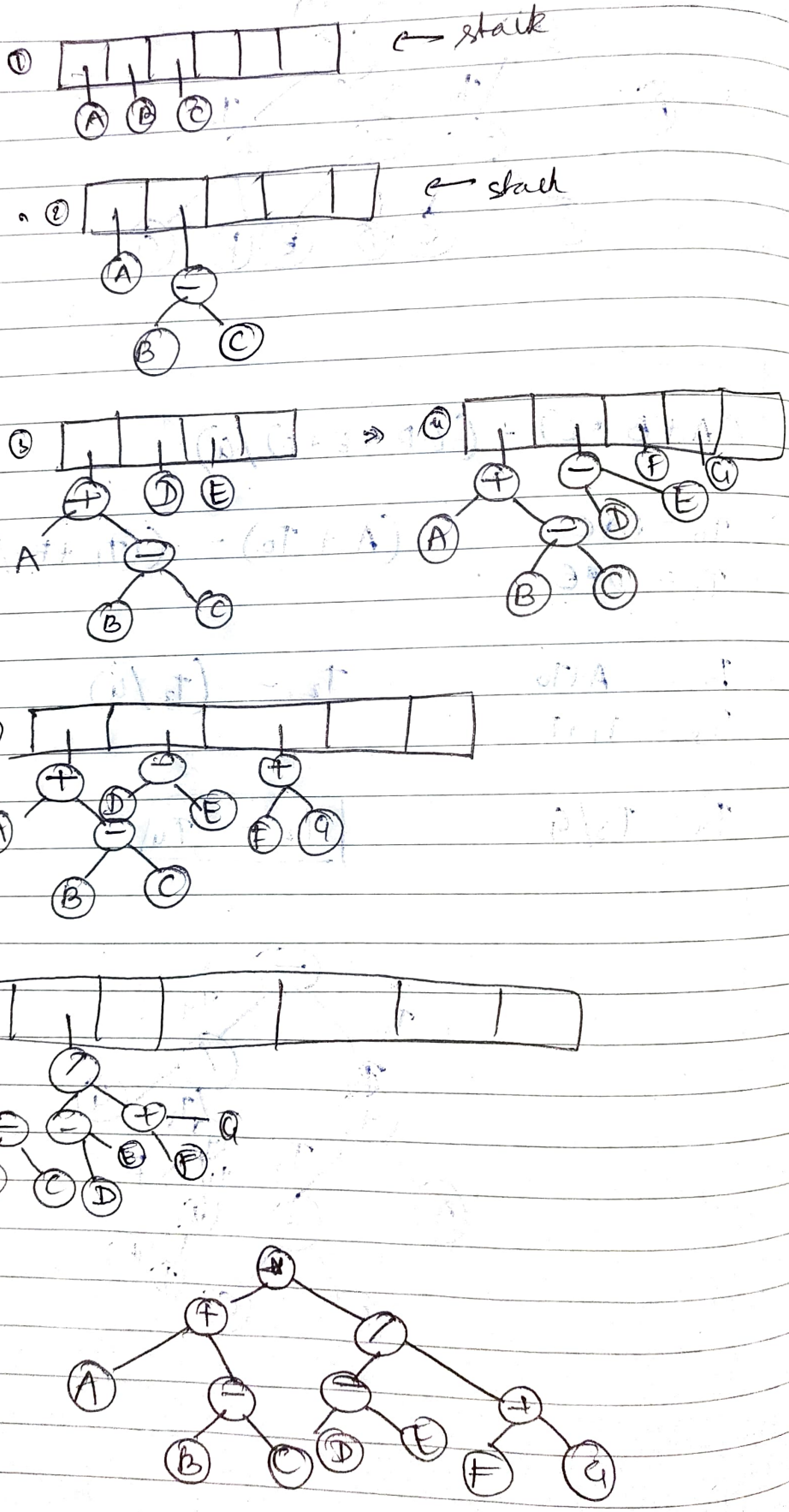
$T_2 = A + T_0$ $\quad T_2 - (T_3 / G)$

$T_3 = T_1 + F$

$T_4 = T_3 / G$ $\quad \boxed{T_2 - T_4}$



① ABC-+DE-FG+/+ [Postorder]



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① ABC - + DE ↑ - FG + / *

a (A + (B - C)) (D ↑ E) (F + G) / *

A + (B - C) * (D - E) / (F + G)

[A + (B - C)] * [(D - E) / (F + G)]

$T_0 = B - C$

$T_1 = D - E$

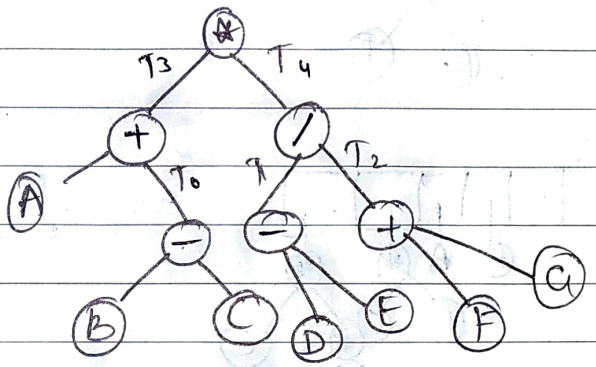
$T_2 = F + G$

$(A + T_0) * (T_1 / T_2)$

$T_3 = A + T_0$

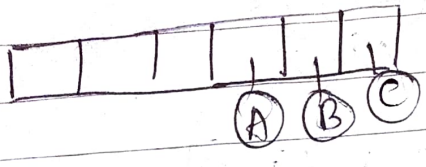
$T_4 = T_1 / T_2$

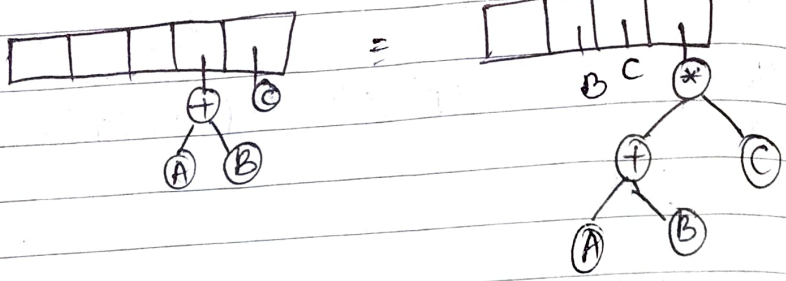
$T_3 * T_4$

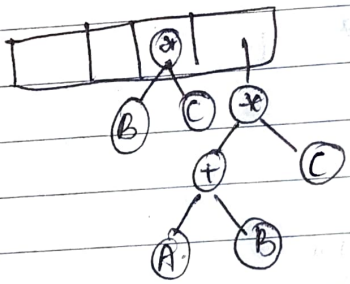


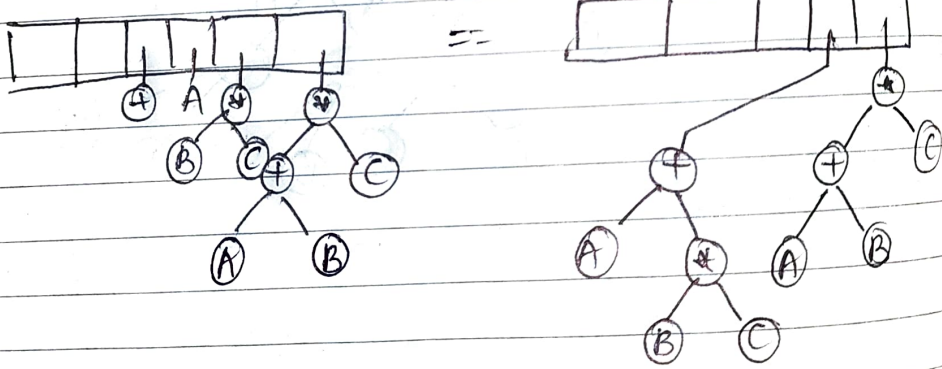

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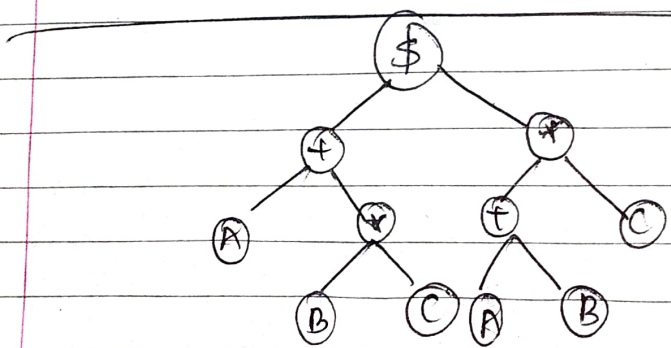
① $+A*BC*+ABC  [Preorder]



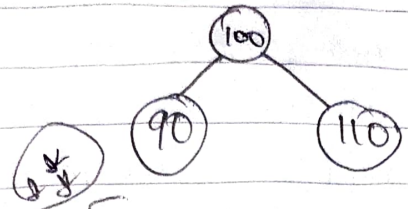



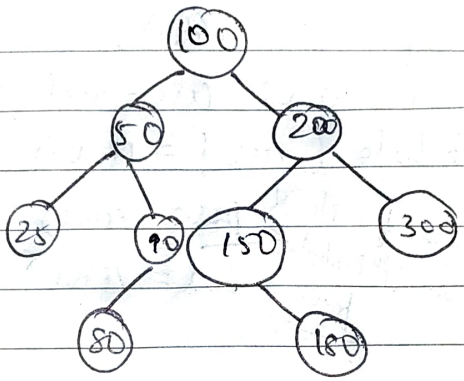

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→ Binary Search Tree: [BST]

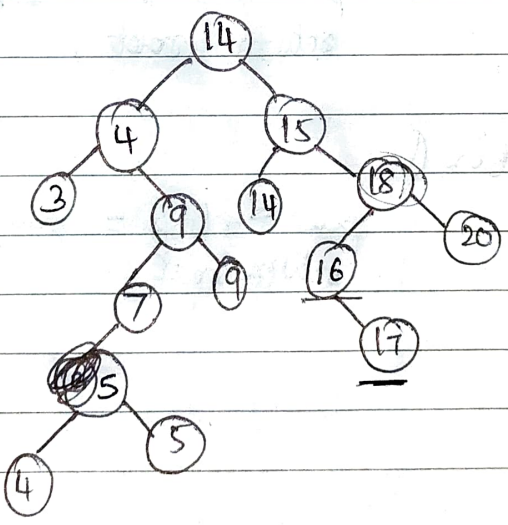


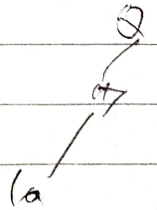

Here, [ Left leaf < ~~Root node~~ Parent node
Right leaf >= Parent node

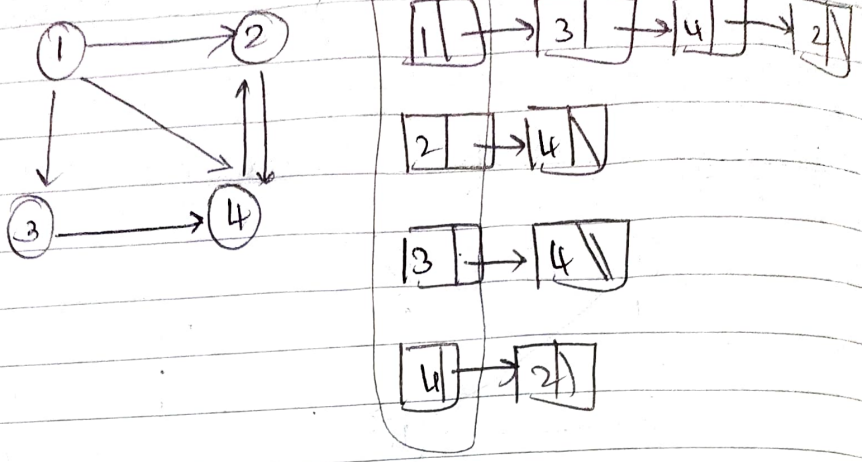
Eg.



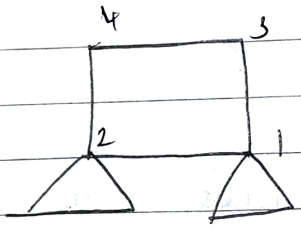

14, 15, 4, 9, 7, 18, 3, 5, 16, 4, 20, 17, 9, 14, 5






→ Atriculation point :-



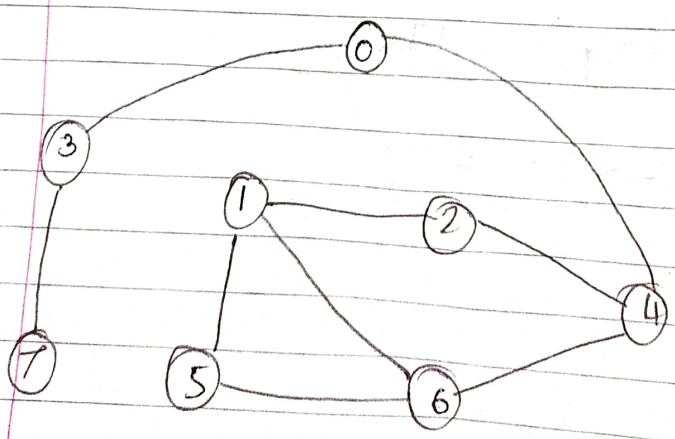

1, 2 → atriculation point
4, 3 → Not

→ Traversal Techiques:

1] Depth first search [DFS]:

2] Breadth first Search [BFS]

① DFS:




| | |
|---|---|
| 0 | 1 |
| 1 | 6 |
| 2 | 5 |
| 3 | 2 |
| 4 | 4 |
| 5 | 7 |
| 6 | 8 |
| 7 | 3 |

5
5
1
2
4
4
X
0

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4 → 2

culation point

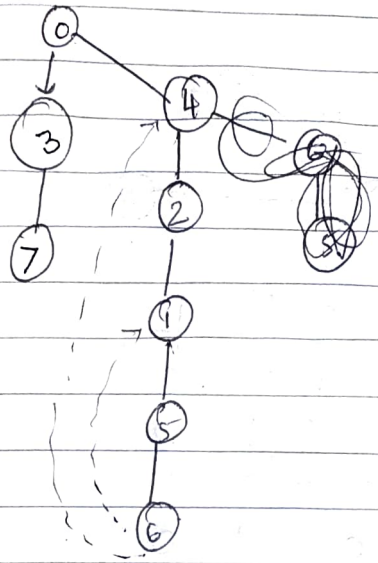
stack

| | |
|---|---|
| i | 6 |
| 6 | 5 |
| 5 | 1 |
| 2 | 2 |
| 4 | 4 |
| 7 | 7 |
| 8 | 3 |
| | 0 |

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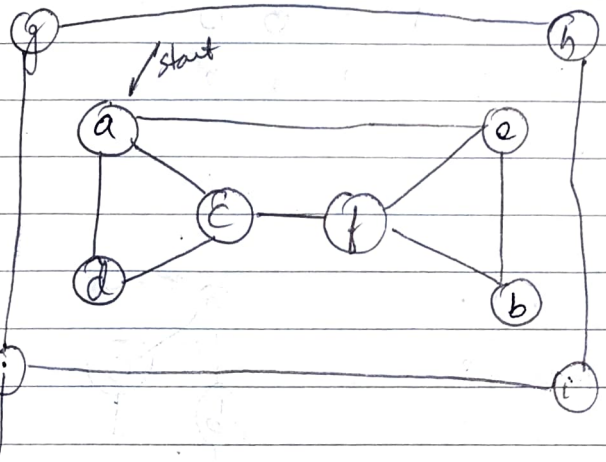


DFS:



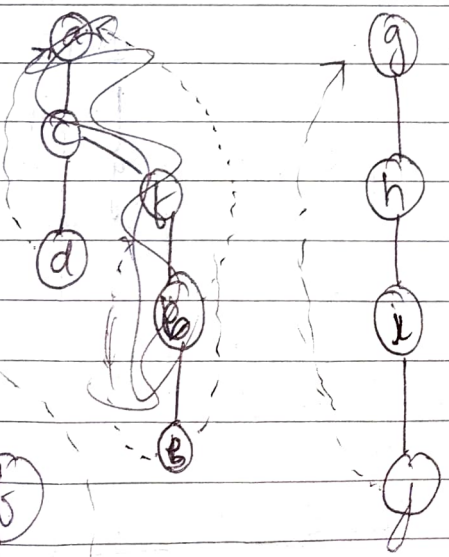

order of visit

0 3 7 4 2 1 5 6




| stack | |
|---|---|
| j | |
| i | |
| h | |
| g | |
| b | |
| e | |
| f | |
| d | |
| c | |
| a | |

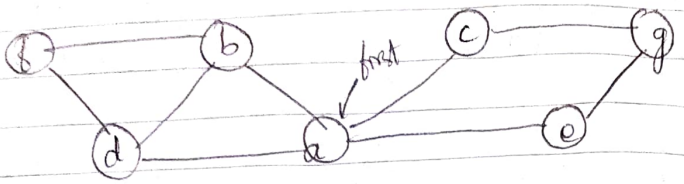
| away | |
|---|---|
| a | 1 |
| b | 5 |
| c | 2 |
| d | 3 |
| e | 6 |
| f | 4 |
| g | 7 |
| h | 8 |
| i | 9 |
| j | 10 |




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full

## Adjacency matrix & graph (DFS)

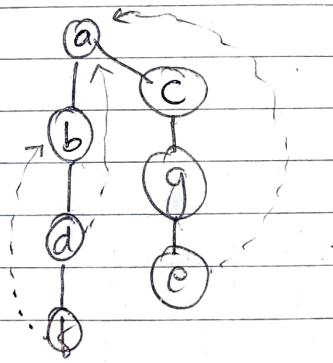



| | a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|---|
| a | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| b | 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| c | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| d | 1 | 1 | 0 | 0 | 0 | 1 | 0 |
| e | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| f | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| g | 0 | 0 | 1 | 0 | 1 | 0 | 0 |

stack

| | |
|---|---|
| e | 1 |
| g | 5 |
| c | 6 |
| f | 1 |
| d | 2 |
| b | 3 |
| a | 7 |

| | |
|---|---|
| a | 1 |
| b | 2 |
| c | 5 |
| d | 3 |
| e | 7 |
| f | 4 |
| g | 6 |




a b d f c g e

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